



NATIONAL UNIVERSITY OF TECHNOLOGY

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Project Progress Report

Early Detection of Social Isolation in Campus Life via Machine Learning

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1 Introduction and Background

1.1 Introduction & Motivation

Social isolation among university students is an escalating issue that severely impacts mental wellness, academic performance, and overall student retention. Transitioning to campus life often strips students of their established support networks, making them vulnerable to loneliness and withdrawal. The primary motivation of this project is to shift campus mental health strategies from reactive to proactive. By analyzing anonymized campus activity logs (such as library visits, dining hall usage, and event participation), we aim to develop a machine learning-based early warning system that detects behavioral changes indicative of social withdrawal.

1.2 Background & Domain Knowledge

This project falls under the domain of **Educational Data Mining (EDM)** and **Behavioral Analytics**. Traditionally, student wellness has been gauged through self-reported surveys or counseling center visits—methods that only capture data after a student is already in distress. Machine learning offers a modern alternative by treating human behavior as time-series data. The core concept relies on anomaly detection and clustering. By establishing a "baseline" of healthy social and academic routines, ML models can flag significant deviations. This system provides a blueprint for proactive intervention, enabling university administration and counseling centers to initiate low-pressure wellness checks before a crisis occurs.

2 Literature Review

Recent research has increasingly focused on the intersection of machine learning and mental health, yet a gap remains in practical, localized applications.

- **Wang et al. (2020) - *StudentLife: Assessing Mental Health via Smartphone Data***: This study demonstrated that passive sensor data from smartphones correlates heavily with student stress and depression. While highly accurate, continuous smartphone tracking raises severe privacy concerns. *Difference*: Our project relies on existing, anonymized campus infrastructure data (ID swipes) rather than invasive personal device monitoring.
- **Chen & Liu (2022) - *Predicting Student Retention Using Campus Facility Usage***: This research utilized dining and library logs to predict dropout rates, using Random Forest classifiers. *Difference*: While Chen focuses strictly on academic retention, our project repurposes similar infrastructural data specifically to measure "Social Density" and emotional wellness, utilizing unsupervised anomaly detection (Isolation Forests).
- **Addressing the Static Data Gap**: Most existing solutions in academic literature test their hypotheses on generalized, static Kaggle datasets. Our proposed solution differentiates itself by generating and analyzing a primary "ground-truth" dataset specific to our university peers, ensuring the model is tailored to our specific campus geography and culture.

3 Preliminary Experiments and Results

Currently, the project is in the data acquisition and exploratory phase. Since we are building a primary dataset, our preliminary experiments have focused on simulating the expected data structure and running baseline models to test our pipeline.

- **Data Simulation & EDA:** We generated a synthetic dataset mimicking the logs of 200 students over a 4-week period using Python (Pandas and NumPy). We engineered a "Social Density" score based on the ratio of communal space usage versus private dorm time.
- **Initial Modeling:** We applied a baseline K-Means Clustering algorithm to segment the simulated student profiles. The model successfully identified three distinct clusters: Highly Social, Academically Focused (high library, low events), and At-Risk (high private dorm time, low dining/communal time).
- **Preliminary Anomaly Detection:** We ran an initial Isolation Forest implementation via Scikit-Learn. When calibrated to a 5% contamination rate, the model successfully flagged the simulated anomalies (students who exhibited a sudden 30% drop in communal activities week-over-week).

4 Problems Faced and Potential Solutions

1. **Problem: Privacy Constraints and Data Collection.** Convincing students to volunteer their daily movement and interaction logs poses a significant challenge due to privacy concerns. **Solution:** We are strictly utilizing anonymized identifiers. Furthermore, we are designing the data collection survey to be as frictionless as possible, relying on aggregated weekly summaries rather than minute-by-minute tracking.
2. **Problem: Class Imbalance.** Social isolation is fundamentally a "minority" class state. Traditional accuracy metrics can be misleading because simply predicting that everyone is "fine" will yield a high accuracy rate. **Solution:** We are shifting our evaluation metrics. Instead of relying on raw accuracy, we will evaluate our models using Precision-Recall (PR) Curves to ensure we minimize false negatives (missing an isolated student) while managing false positives.
3. **Problem: Capturing Temporal Behavioral Changes.** A static snapshot of a student's data does not account for natural exam-week exhaustion versus genuine social withdrawal. **Solution:** If standard clustering fails to account for time-based changes, we plan to transition from Scikit-Learn to PyTorch to implement Long Short-Term Memory (LSTM) networks, which are explicitly designed for sequential data analysis.

5 Objectives Met and Yet to be Met

5.1 Objectives Met

- Formulated the core problem statement and finalized the project proposal.
- Completed the literature review and identified the technical gap (primary vs. secondary data).

- Designed the data collection framework and the parameters for the longitudinal survey.
- Set up the Python environment, established the data preprocessing pipeline, and successfully tested synthetic data on K-Means and Isolation Forest algorithms.

5.2 Objectives Yet to be Met

- Deploy the longitudinal survey to gather actual peer "ground-truth" data over the designated time frame.
- Perform rigorous feature engineering on the real-world dataset to finalize the "Social Density" metric.
- Train the final classification and anomaly detection models on the real dataset.
- Evaluate the model using Mean Absolute Error (MAE) and Silhouette Coefficients, and compile the final analytical report.

6 Conclusion

The project is progressing steadily according to the initial proposal timeline. The transition from defining the scope to establishing a functional data pipeline has been successful. Preliminary tests on synthetic data confirm that algorithms like K-Means and Isolation Forests are well-suited for identifying behavioral clusters and anomalies in campus activity data. The primary focus for the next phase will be the ethical collection of live survey data and addressing the temporal complexities of student habits. Once completed, this model has the potential to serve as a foundational tool for proactive student welfare management on campus.